

Amos Hebb

Radar and machine-learning engineer taking RF-sensing research to shipped systems.

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EDUCATION

Delft University of Technology

PhD Candidate in Electrical Engineering

Waveform diversity for ground surveillance radar, under Prof. Francesco Fioranelli. Learning transmit waveforms from a radar's own returns on PARSAX, a software-defined dual-polarimetric radar.

Delft, Netherlands

Sep 2025 – Present

University of Toronto

Master of Applied Science in Computer Engineering

Reinforcement learning for radar resource management, under Prof. Raviraj Adve.

Toronto, Canada

Sep 2022 – Sep 2024

Royal Military College of Canada

Bachelor of Engineering in Computer Engineering

First Class Distinction.

Kingston, Canada

Aug 2015 – May 2019

EXPERIENCE

Senior Consultant

Deloitte – Omnia AI

- Led a 6-developer team building Health Canada's proof-of-vaccination "Issuer in a Box", the certificate-generator service behind Canada's national vaccine passport (Node, TypeScript)
- Built the IMCO cyber-risk dashboard (SQL, Python) and PSPC real-property portfolio optimization models (Gurobi, Python) for federal clients

Dec 2021 – Sep 2022

Toronto, Canada

Data Scientist

HSBC Global Banking and Markets

- Shipped production chained binary classifiers with SHAP explanations over petabyte-scale trading data (PySpark, Python)
- Built graph-based counterparty record linkage (Spark, GraphX) and led an intern developing a named-entity-recognition model (SparkNLP)
- Re-engineered a manual Excel and Python reporting process into a petabyte-scale ETL pipeline (Hadoop, Spark, Scala)

Jul 2019 – Dec 2021

Toronto, Canada

Aerospace Engineering Officer

Royal Canadian Air Force

- Deployed to the far north on Operation Nanook 2018 with 405 Long Range Patrol Squadron
- First-line maintenance training with 14 Air Maintenance Squadron; search-and-rescue training with 413 Transport and Rescue Squadron

Jun 2015 – Jul 2019

Greenwood, Canada

PROJECTS

Belief-Reward Radar Policies *Reinforcement Learning, Julia*

- Particle-filter multi-target tracking and a Belief-MCTS solver for a radar POMDP; abstract accepted to IEEE RadarConf24

Dec 2023 – May 2024

RL for Multi-Task Radar Scheduling *Gymnasium, Python*

- Gymnasium simulation of a task-queue scheduler, porting MATLAB algorithms to Python and training reinforcement-learning agents against it

Sep 2023 – Dec 2023

Depth from Mono Capstone *ROS, OpenCV, SLAM*

- ROS-based turtlebot with a convolutional network trained to estimate LIDAR depth from a single camera

Aug 2018 – Jun 2019

TECHNICAL SKILLS

Languages

Native English, CBC French

Programming

Python, Julia, TypeScript, with Go and Rust for fun

Robotics

ROS, ArduPilot, Donkey Car

Other

Photography, first job was as a photo-lab technician